



Deception in avatar-mediated virtual environment

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ABSTRACT

This study explored the effects of avatars on deception – how perceived avatar likeness to self can affect the truthfulness and accuracy of interactions online. More specifically, this study examined the extent to which perceived avatar similarity influences self-awareness and users' degree of attraction to them, and how these psychological states affect deception in the context of Second Life. The results, based on web-based survey data of 159 Second Life users, revealed that avatar similarity in attitude and behavior to the owner heightened self-awareness, which, in turn, reduced deception. Perceived avatar similarity in terms of appearance was found to have a direct negative impact on deception so that those who perceived their avatars to look similar to themselves were less likely to engage in deceptive behavior. Implications of the findings are discussed.

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1. Introduction

With the proliferation of more online spaces in which to interact, the use of avatars has become more prevalent. In recent years, avatars have emerged to give an added dimension to Computer-Mediated Communication (CMC). Avatar is defined as a digital representation through which users live and relate to others in the virtual world. They represent our virtual bodies or virtual selves, which greatly change how we interact with others both in avatar-based online communities (e.g., virtual worlds or Massively Multiplayer Online Role-Playing Games [MMORPGs]) as well as subsequent face-to-face interactions (Yee, Bailenson, & Ducheneaut, 2009).

Previous studies have revealed that the use of avatars can bring about feelings of co-presence (Garau et al., 2003), satisfaction (Holzwarth, Janiszewski, & Neumann, 2006), and even increase perceptions of credibility of one's communication partner (Nowak & Rauh, 2005). They have also been found to affect their users in the real world to the extent that people behave like they possess the traits of their avatars (Yee et al., 2009). In addition, the use of avatars that portray their users' physical characteristics and personal preferences have been reported to act like a self-reflecting mirror, increasing self-awareness (Vasalou, Joinson, & Pitt, 2007).

This study explores the effects of avatars on deception, which has received relatively little scholarly attention. Deception is a common occurrence in everyday personal life (Hancock, Thomas-Santelli, & Ritchie, 2004; Zhou, Burgoon, & Twitchell, 2003) and in business (Cialdini, Petrova, & Goldstein, 2004; Zuckerman,

DePaulo, & Rosenthal, 1981). It may refer to the concealment of identity or the transmission of false information, and in this study, it refers to telling untruths of information about oneself (e.g., one's appearance, interests and personal information).

Research suggest deception to be more prevalent in virtual environments because the relative anonymity provided by CMC offers more opportunities for manipulating self-presentation (Walther & Tidwell, 1996). In CMC, interpersonal and audio-visual cues that may give one away (e.g., disfluencies or nervousness) are also typically eliminated, making it easier for people to deceive online. The use of avatars in virtual environments can further foster deception due to the ease of switching identities or manipulating multiple identities. Avatars can be used as digital disguises (like wearing a “mask”), which lead to reduced anxieties when deceiving others (Galanxhi & Nah, 2007). As computer-mediated environments make it easy to deceive, it is important to study these deceptive behaviors. With a better understanding of deception in CMC, not only can users of virtual environments be educated about online deception, but also designers may be informed of the potential of avatars as an avenue to create trustful environments that support interpersonal bonds. Given that people tend to create avatars that have some similarities with themselves (Messinger et al., 2008), we aim to explore how self-similar avatars affect deception. In the context of online communication where inaccurate presentation of the self is easy and untraceable to one's real identity, it is important to understand how avatars can affect identity play and influence the veracity of interaction, lending significance to this study.

Using theories of self-awareness (Duval & Wicklund, 1972) and similarity-attraction (Byrne, 1971), we specify the mechanisms through which avatar affects deception in virtual environments. More specifically, we focused on several unexplored theoretical

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factors, such as self-awareness, similarity perception and attraction, and examined the ways in which these factors mediate the impacts of avatar on deception. There is a dearth of research on the effects avatars have on deception, and the limited studies that exist mostly linked deception with the perceived anonymity in virtual environments (e.g., Donath, 1999; Walther & Tidwell, 1996). However, research suggests that deception can be mediated by other psychological states, such as self-awareness (Lewis, Stanger, & Sullivan, 1989; Mazar, Amir, & Ariely, 2008; Mazar & Ariely, 2006). Self-awareness is the ability to shift attention from the environment to the self and vice versa (Govern & Marsch, 2001). It can result in honest behavior that is in accord with social and personal standards (Beaman, Klentz, Diener, & Svanum, 1979; Mazar et al., 2008), but it may lead to greater ability to deceive as well (Johnson et al., 2005). At the same time, other factors, such as attraction, vis-à-vis the Proteus effect (Yee et al., 2009), may similarly mediate deception. Attraction may lead to honesty when one wears an avatar that one likes and feels confident in. However, when one is attracted to one's avatar, one may aspire to be like the avatar and deceive others about one's real self, particularly in physical aspects. It is therefore of interest to examine how self-similar avatars would affect deception through these alternative processes. In doing so, this study can extend current knowledge on the influence wielded by avatars in affecting behavior, by revealing different paths through which avatar use impacts deception.

The study used a web-based survey administered to 159 users of Second Life, a virtual world that allows people to socialize, as well as create and trade virtual property. Second Life gives great freedom to users to create and customize avatars, making it ideal to explore avatar effects. Using structural equation modeling (SEM), we examined our research model and tested research hypotheses. The findings and their implications are discussed.

2. Literature review

2.1. Deception in Computer-Mediated Communication

Deception refers to the active transmission of messages and information to bring about a false conclusion (Buller & Burgoon, 1996). People lie almost daily (DePaulo, Kirkendol, Kashy, Wyer, & Epstein, 1996) and there are many ways to deceive, such as lies, fabrications, evasions, concealments, misdirection, bluffs, fakery, mimicry, tall tales, white lies, deflections, evasions, equivocation, exaggerations, camouflage and strategic ambiguity (Burgoon & Nunamaker, 2004). Generally, deception can be related to identity or the content of the message. While some are clearly deceptions aimed at providing a false impression or misleading information, others may be some form of identity manipulation (Galanxhi & Nah, 2007). Deception in interactive communication is a dynamic rather than static event and deceivers can modify and adapt their communicative performance to changing circumstances (Burgoon, Stoner, Bonito, & Dunbar, 2003).

In everyday life, people commonly lie about their feelings, actions, plans, their whereabouts, their knowledge and achievements (DePaulo et al., 1996). Evidently, lies are mainly told to benefit the self (DePaulo et al., 1996). They may be told to align oneself to expectations of behavior or responses (Millar & Tesser, 1988; Yeung, Levine, & Nishiyama, 1999) or to obtain social approval through conformity (Froming, 1978; Zerbe & Paulhus, 1987). Lies told to benefit others are usually to protect them from embarrassment, worry or hurt. They may boost others' self-confidence (Vrij, 2000) so that they feel better about themselves.

These may be reasons for online deception as well though studies have shown that given the wider scope in which people can lie, motivations for online deception are much broader, including role

play, exploration of alternate identities and privacy concerns (Caspi & Gorsky, 2006; Joinson & Dietz-Uhler, 2002; Whitty, 2002). Deception in the form of misrepresentation (Cornwell & Lundgren, 2001; Ellison, Heino, & Gibbs, 2006; Hancock, 2007; Toma, Hancock, & Ellison, 2008; Vrij & Holland, 1998), presentation of an ideal self (McKenna, 2007; McKenna, Green, & Gleason, 2002; Turkle, 1995a; Whitty, 2008) or category deception (e.g., gender-swapping; Boellstorff, 2008; Bruckman, 1996; Hussain & Griffiths, 2008; Turkle, 1994) are common as well. Some of such lies can be traced to impression management concerns, which are high despite the mediated channel. People not only lie about their looks and physical attributes, using false descriptors or photographs of another, but also their socio-economic status (Gibbs, Ellison, & Heino, 2006; Whitty, 2002). These ego-inflating enhancements may be characteristics that one finds attractive and aspires to have in real life, or the expression of an alternate identity.

2.2. Effect of avatar on deception

2.2.1. Avatar similarity

Turkle (1995b) has proposed the notion of people creating multiple online selves (or "a distributed self") with new online identities (p. 118). For instance, people live multiple virtual lives through their avatars (e.g., as a vampire or a warrior) just as they play different roles in real life (e.g., as a hockey player, a lawyer and a son). These virtual lives may be vastly different – not only in outward representations of appearance but also attitude, manner and behavior – from one another and from real life. It is therefore possible to evaluate the similarities of these different selves with one another and with the real self.

Similarity is an important factor that influences attraction (Byrne, 1961a, 1961b; Byrne & Griffitt, 1969), intimacy (Griffin & Sparks, 1990; Heller & Wood, 1998), and social judgments (Lambert, 1995) in various interpersonal and group settings. According to the similarity–attraction hypothesis (Byrne, 1971), the more similar a person is to oneself, the greater the attraction for that person. The similarity serves as reinforcement and validation that one's concepts and precepts are in line with others' and constitute a rewarding interaction (Byrne, 1971). Self-categorization theory also suggests that because individuals desire a positive self-identity they tend to evaluate more positively those that belong to the same social category as them (Tsui, Egan, & O'Reilly, 1992).

Previous studies have examined similarity in terms of personality (Byrne, Griffitt, & Stefaniak, 1967; Strauss, Barrick, & Connerley, 2001), attitude, (Byrne, 1961a, 1961b; Byrne et al., 1967), ideal self (Herbst, Gaertner, & Insko, 2003), as well as appearance (Crosby, Evans, & Cowles, 1990). A term often used is homophily, which refers to the similarity between people with respect to certain attributes, like beliefs, values or social status (Rogers & Bhowmik, 1970). Since different types of attribute may affect deception in a different manner, homophily can be examined separately as comprising the psychological (i.e. beliefs and attitudes) and tangible (i.e. appearance) aspects.

Homophily can be conceptualized as subjective (perceived) or objective. Perceived homophily is the degree to which a member of a dyad perceives the other as being similar in terms of attributes while objective homophily is the degree of observable similarities between the two (Rogers & Bhowmik, 1970). Both perceived and objective homophily have been used in a number of studies, producing meaningful results (McCroskey, Richmond, & Daly, 1975).

Previous research indicates that perceived, rather than actual homophily has a stronger effect. Perceived personality homophily, for instance, is associated with greater friendship intensity than actual similarity in personality (Selfhout, Denissen, Branje, & Meeus, 2009). In research on performance appraisal, higher subordinate performance evaluations by superiors has generally

been associated with perceived, rather than actual, similarity (Strauss et al., 2001; Turban & Jones, 1988). The resultant effect of perceived similarity is due to people reacting on “the basis of perceptions of reality, not reality per se” (Ferris & Judge, 1991, p. 464). Apparently, believing that others believe what you believe fulfills the need for consistency (Montoya, Horton, & Kirchner, 2008). What is important then is that individuals believe that the other is similar to them, whether or not there is actual similarity (Condon & Crano, 1988; Morry, 2005).

Researchers have applied the same line of arguments to avatar-self similarity. Perceived avatar similarity has a number of effects on users, including heightened self-awareness (Vasalou et al., 2007) and greater self-presence (Ratan, Santa Cruz, & Vorderer, 2008). People with representative avatars also identify more with their avatars, which positively affect their game enjoyment (Trepte & Reinecke, 2010). In the present study, we explore how perceived avatar similarity affects self-awareness, which in turn influences deception in computer-mediated environments.

2.2.2. The impact of avatar similarity on self-awareness

The concept of self-awareness is dependent on the orientation of conscious attention. It is also understood to be self-focused attention or selective processing of information about the self. Self-awareness theory states that when self-awareness becomes salient, people focus their attention on themselves. Studies have shown that environmental and external stimuli, such as mirrors, an audience, or being videotaped or recorded can direct people's attention towards the self, leading to the salience of public or private aspects of the self (Carver & Scheier, 1978; Davis & Brock, 1975; Geller & Shaver, 1976; Wicklund & Duval, 1971). For instance, a mirror gives rise to behaviors that show an awareness of the covert, personal aspects of the self (Baldwin & Holmes, 1987; Froming, Walker, & Lopyan, 1982; Scheier & Carver, 1977, 1980). This makes people self-conscious as objective evaluators of themselves, and they will appraise and compare their current behavior to their internal standards and values. As such, people are more likely to align their behavior with their internal moral standards, such as honesty. A television camera, an audience or a tape recording of one's voice can elicit behaviors that show an awareness of the overtly displayed and social aspects of the self. Thus made self-aware, people attend to the social aspects of themselves and are concerned with the impression that they convey to others through their behavior (Webb, Marsh, Schneiderman, & Davis, 1989).

In the context of avatars, people have the choice to make them self-resembling or not, and the extent an avatar is perceived to be similar to its owner can affect self-awareness. The perception of similarity may have the effect of focusing attention towards the private aspects of the self, much like looking into a mirror and seeing a self-image, as Vasalou et al. (2007) suggest in their study. They asked subjects to create avatars, giving them free rein to create the kind of avatar they want, then had them answer a questionnaire measuring self-awareness and perceived avatar similarity. They found a significant correlation between avatar similarity and private self-awareness, and suggest that users who customized their avatars to look like themselves experienced increased self-awareness compared to those who created avatars different from themselves because they perceive their avatar as resembling their self-reflection in a mirror.

2.2.3. Avatar similarity and its impact on deception via self-awareness

Empirical studies have produced mixed findings about the effect of self-awareness on deception. On the one hand, it has been argued that self-awareness is associated with a greater ability to deceive (Johnson et al., 2005) as self-awareness is linked with greater introspection (Beaman et al., 1979). Since self-awareness is closely tied to understanding one's own thoughts, it is assumed

to “give rise to the ability to reflect on the thoughts of others” (Keenan, Gallup, & Falk, 2003, p. 78). This mental state attribution or theory of mind is useful in deception to determine what another individual in a similar situation is thinking. Johnson et al. (2005) have shown that individuals with high private self-awareness are better deceivers.

On the other hand, other studies have shown that private self-awareness is associated with an adherence to personal standards of behavior such as honesty (Beaman et al., 1979; Mazar et al., 2008). Faced with a conflict between situational standards (i.e. behavior or response expected in an environment) and personal standards, high private self-awareness is associated with adherence to own values and beliefs with less concern for social expectations or conformity pressures (Fenigstein, 2009; Froming & Carver, 1981). Thus, self-aware individuals would emphasize self-standards and not moderate their views or lie about them in order to conform to others' differing opinion. It is this tendency to act in accordance with internal, personal yardstick that standards, like honesty, are adhered to because people become attuned to the discrepancies between their actual and ideal selves. They experience discomfort and attempt to eliminate or reduce them through matching behavior with the standards (Gibbons & Wicklund, 1982), thus reducing instances of deception.

In sum, the aforementioned studies suggest that avatar similarity can be a stimulus that can direct users' self-awareness. Although previous studies have produced mixed findings with regard to the effect of self-awareness on deception, a majority of empirical studies have shown that self-awareness decreases deception as cognizance of internal values and standards is increased. Therefore, the study posits that:

H1. Avatar similarity will be positively associated with self-awareness.

H2. Self-awareness will be negatively associated with deceptive behavior.

H3. Avatar similarity will influence deception through its impact on self-awareness.

2.2.4. Avatar similarity and attraction

Attraction refers to attitudinal positivity (Huston & Levinger, 1978). Researchers (e.g., McCroskey & McCain, 1974) have distinguished between liking and respect as different dimensions of attraction. They argued that attraction is a multi-dimensional construct, constituting physical, social or task attraction. Physical attraction relates to the physical features and dress, social attraction to the liking of the other as a friend and task attraction is analogous to respect, or how easy and worthwhile working with the other would be (McCroskey & McCain, 1974). For instance, while physical and social attraction evaluate the avatar in terms of how it looks and how much one would like it based on its social qualities, task attraction appraises the avatar on the extent it can be relied on for tasks and how enjoyable the experience of working with it would be.

Studies on similarity attraction assert that people seek out and are attracted to those who are similar to them because being in agreement with others satisfy the self-evaluative drive or validation of their beliefs and attitudes. Perceived similarity in psychological aspects, like attitude (Rocca & McCroskey, 1999; Rosenblatt & Greenberg, 1988; Sunnafrank, 1986) and personality (Hansson & Fiedler, 1973; Lerner & Agar, 1972), is linked to attraction. This similarity–attraction appeal extends to facial similarity. People are attracted to faces similar to their own due to repeated

exposure to their own and other genetically similar faces, like family or relatives (Hinsz, 1989). Research has shown that people are inclined to see other people who are similar to themselves in appearance, beliefs or personality not only as more attractive, but also as more persuasive and appealing (Bailenson, Garland, Iyengar, & Yee, 2006; Byrne, 1971). Due to the shared attributes – in terms of appearance or attitudes – those who are similar have some basis to identify with each other. The strands of similarity are believed to strengthen the bond between people (Bailenson et al., 2006; Brock, 1965).

Taken together, it is predicted that an avatar that is perceived to closely resemble its owner may generate in the owner a greater liking for it, just as between humans. If avatars share the same attitudes, values and beliefs as their owners, they would be perceived by their owners as being more like themselves. As such, physical attraction would be high since people prefer those who share some physical similarities, e.g., facial features (Bailenson, Iyengar, Yee, & Collins, 2008; Hinsz, 1989). Similarly, social attraction for avatar would be high, as evidenced by the high social attraction between dyads matching in communication traits (e.g., temperament and communication styles; Martin & Anderson, 1995). There would also be high social and task attraction for the avatar because like homophilous dyads, they are reinforcing (Senn, 1971), with little need for constant validation (Burlison, Kunkel, & Szolwinski, 1997). We therefore expect physical, social and task attraction with homophilous owner–avatar pairs.

It is posited that:

H4. Avatar similarity will be positively associated with attraction to avatar.

2.2.5. Avatar similarity and its impact on deception via attraction

Self-perception theory states that people can infer their own attitudes, beliefs and emotions from self-observation of their behavior (Bem, 1972). Uniforms and costumes can act as cues for behavior (Johnson & Downing, 1979), for instance, people who don black uniforms see themselves as aggressive, tough and mean, much like what observers perceive of them (Frank & Gilovich, 1988). In the same vein, people may “make inferences about their expected dispositions from their avatar’s appearance and then conform to the expected attitudes and behavior” (Yee et al., 2009, pp. 293–294), a phenomenon referred to as the *Proteus effect*. Avatars, and even the uniforms or costumes they wear, can alter people’s behavior and attitudes (Bruckman, 1996; Peña, Hancock, & Merola, 2009; Suler, 1996; Yee et al., 2009). People with taller avatars, for instance, behaved more aggressively because the avatar primes associated qualities of self-esteem and competence while those with more attractive avatars walked closer to and disclosed more personal information to their partners because attractiveness is associated with extraversion and friendliness (Yee et al., 2009). In the same way, people who are attracted to their avatars might be imbued with the confidence associated with their represented selves. Since more confident individuals are less inclined to lie (Vrij & Holland, 1998), we expect reduced deception. This leads to the next hypothesis:

H5. Attraction to avatar will be negatively associated with deception.

Taken together, the literature suggests that avatar similarity can lead to attraction for the avatar, which, in turn, can influence deception. Thus, the study posits:

H6. Avatar similarity will influence deception through its impact on attraction.

3. Method

3.1. Sample and data collection

We used survey research to explore the relation between avatar and deception, given its advantage in reflecting deception in natural contexts (Miller & Stiff, 1993). The deployment of a web-based survey can aid in reducing impression management (Booth-Kewley, Edwards, & Rosenfeld, 1992) and responding in socially desirable ways (Kiesler & Sproull, 1986; Sproull & Kiesler, 1991), with more truthful reports produced (Sproull & Kiesler, 1991). It can mitigate, to some extent, the need for social desirability, one of the main reasons for deception online. Survey methodology has been used in a number of studies involving deception, for example, to investigate misrepresentation in online and offline situations (Cornwell & Lundgren, 2001), dishonesty in chatrooms (Whitty, 2002) and frequency of deceptive behavior (Lewis & George, 2008).

We collected data for this study from users of Second Life, an online 3D virtual world with content that is constructed and created by its users. With a myriad of tools to build and customizations to control the appearance of one’s avatar, from hair color to foot size, Second Life is suited to investigate the effect of identity creation and performance since people can choose to be what they want to be and create a self that is totally different from their real selves.

Respondents were recruited in Second Life and they accessed a webpage to complete the survey, which took approximately 20 min. Convenience sampling was utilized since it was impossible to obtain a comprehensive list of all Second Life users from which we can draw random samples. Avatars at a variety of entertainment, educational and shopping locations in Second Life were invited to participate. As well, recruitment was conducted using Second Life advertising and music-related groups. A token of 100 dollars of game currency (i.e. 100 Lindens or the equivalent of 50 cents) was given upon completion of the survey. A total of 246 people responded to the survey and 159 fully completed surveys were received. Since the survey asked about sensitive topics, little demographic information was collected to reduce deception. At the time of data collection, Second Life required users to be at least 18 years old, therefore we can assume that respondents were aged 18 or older. Respondents were mainly from the United States (29%), with Asians making up 23%, Middle-easterners 17%, Western Europeans 16%, and Australians, Scandinavians and Latin Americans constituting the rest.

3.2. Measures

For all the measures, the variables were measured with new scales adapted from scales validated in previous studies. All items used, descriptive statistics and reliability of scales are shown in Table 1.

3.2.1. Avatar similarity

Avatar similarity, referring to the degree of resemblance between a user and the avatar, was operationalized as physical similarity and homophily. Since perceived similarity was measured, self-reports were used. To assess the degree of physical similarity, we based our 5-item scale on Messinger et al.’s (2008) study. Homophily was conceptualized as the degree of similarity between the self and the avatar in terms of their attitudes and behavior. It was measured with 5 items adapted from McCroskey, McCroskey, and Richmond’s Measurement of Homophily (2006).

3.2.2. Self-awareness

We adapted Govern and Marsch’s (2001) Situational Self-awareness scale to assess the degree to which people were

Table 1

Survey items employed to construct variables.

Scale	α	Mean	SD	Items
Appearance similarity	.82	3.04	.91	To what extent is your avatar in Second Life similar to your real self in the following dimensions? • Body • Style of clothes • Weight • Height • Overall look
Homophily	.88	1.76	.67	My avatar in Second Life shares my values. My avatar in Second Life treats people like I do. My avatar in Second Life behaves like me. My avatar in Second Life has thoughts and ideas that are similar to mine. My avatar in Second Life has a lot in common with me.
Self-awareness	.77	2.15	.67	When I am using my avatar in Second Life, I am conscious of my inner feelings. When I am using my avatar in Second Life, I am reflective about my life. When I am using my avatar in Second Life, I am aware of my innermost thoughts.
Social attraction	.80	1.92	.74	I think my avatar in Second Life could be a friend of mine. My avatar in Second Life would be pleasant to be with. I could become close friends with my Second Life avatar.
Physical attraction	.92	1.85	.74	I think my avatar in Second Life is handsome/pretty. My avatar in Second Life is sexy looking. I find my Second Life avatar attractive physically. My avatar in Second Life looks appealing. My avatar in Second Life is nice looking. My avatar in Second Life has an attractive face. My avatar in Second Life is good looking.
Task attraction	.63	2.11	.85	I would enjoy working on a task with my Second Life avatar. I would recommend my Second Life avatar as a work partner.
Deception	.90	1.57	.74	While speaking to a stranger in Second Life, have you lied about: • Your age (somewhat older or younger than you really are)? • Where you live? • Your education? • Your job? • Your income? • Your interests (e.g., hobbies, religious orientation, musical preferences)? • Your physical appearance (e.g., hair color, weight, state of health)? • Your marital status (i.e. single, married, separated)?

conscious of themselves while using their avatar in Second Life. This 3-item scale was used instead of the more popular [Fenigstein, Scheier and Buss \(1975\)](#) self-consciousness scale, as the latter was developed to measure state self-awareness, and may not be suitable for our needs.

3.2.3. Attraction

We adapted [McCroskey and McCain's \(1974\)](#) scale to assess the degree to which a user was attracted to his/her avatar. The final scale comprised three sub-dimensions of attraction – social, physical, and task.

3.2.4. Deception

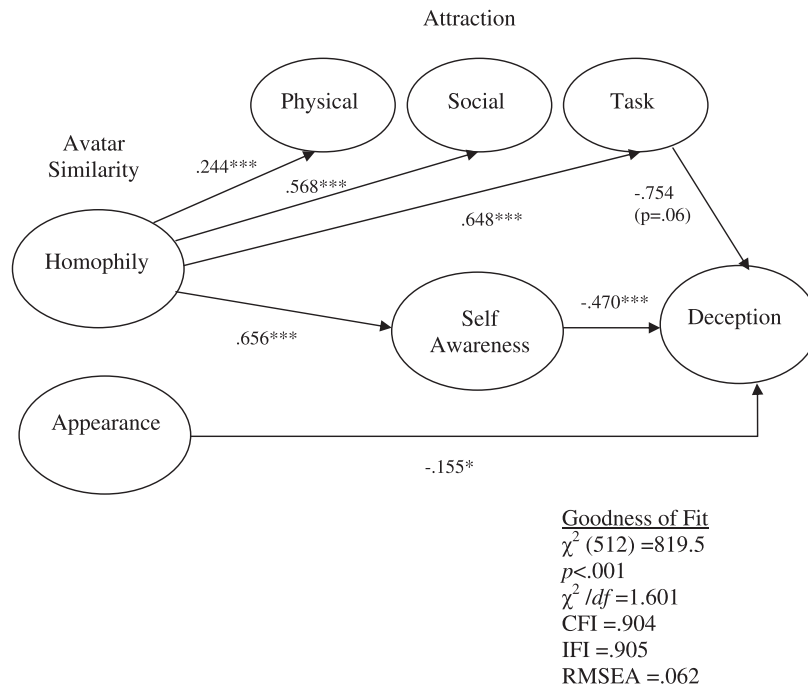
Deception was assessed using a scale based on a study by [Lewis and George \(2008\)](#), who in turn adopted them from scales developed by [Cornwell and Lundgren \(2001\)](#), and [Whitty \(2002\)](#). Word-ing of the 8-item scale was changed to make it suitable for use in Second Life.

4. Results

We analyzed the data collected using SEM, which allows researchers to perform path-analytic modeling with latent variables. The software package we used was AMOS 16, a covariance-based SEM tool. The model included avatar similarity (physical appearance and homophily), attraction (physical, social, and task) and self-awareness as exogenous variables, and deception as an endogenous variable. The first step in the model testing was to

examine the research model's goodness of fit and find evidence for the validity of the measurement model. The results of the initial SEM analysis showed that the chi-square statistics was significant for the global model ($p < .001$). We also found that this model had an inadequate fit as indicated by various alternative goodness-of-fit indices, such as the comparative fit index (CFI $< .90$), the incremental fit index (IFI $< .90$), and the root mean square error of approximation (RMSEA $> .08$).

In short, the results suggested that the research model should be modified in the following manner. The initial results showed that the paths from homophily to self-awareness and to attraction were significant ($p < .001$) as hypothesized, but those from appearance similarity to the two factors were not significant ($p > .05$). Modification indices suggested that appearance similarity could have a direct impact on deception. The direct path may represent the impact of appearance similarity on deception through other mechanisms unexplored in this study, such as impact via perceived anonymity ([Kiesler, Siegel, & McGuire, 1984](#); [Walther & Tidwell, 1996](#)). In general, models can be trimmed or built based on theoretical and/or empirical standards ([Kline, 2004](#)). Therefore, we added a direct path from appearance similarity to deception instead of indirect paths from appearance similarity to deception via self-awareness and attraction. This model specification significantly improved our final model. The results of SEM analysis showed that the chi-square statistics remained significant for the global model ($\chi^2(512) = 819.5, p < .001$), indicating that the fit of data to the hypothesized model was not entirely adequate. However, the appropriateness of chi-square testing for model fitting is routinely questioned owing to its sensitivity to sample size



Note: All parameter estimates are standardized.

* $p < .05$, ** $p < .01$, *** $p < .001$

Fig. 1. Results of SEM analysis for deception model.

and model complexity (e.g., Byrne, 2001). As such, many researchers believe that the significance of the chi-square test may be discounted when a good model fit is indicated by other alternative tests (e.g., CFI, IFI, and RMSEA; Byrne, 2001; Hair, Anderson, Tatham, & Black, 1998). As Fig. 1 shows, the results of SEM analysis revealed an acceptable fit for the revised research model. More specifically, the CFI, IFI, RMSEA, and ratio of chi-square to degrees of freedom were all within acceptable thresholds of goodness of fit. The .062 RMSEA was slightly higher than the suggested .05 benchmark (Byrne, 2001) but below the cut-off level of .08 recommended by Browne and Cudeck (1993). Note that all items in the latent factors exhibited factor loadings above the .50 benchmark (Hair et al., 1998) and had significant p values at the .001 level, thus affirming the construct validity of the measurement model.

Fig. 1 provides the results of model testing for deception. All the paths shown in the revised research model were significant at either the .001 or .05 level. As shown, homophily had a positive effect on self-awareness ($\beta = .656$, $p < .001$). Hence, H1 is partially supported. Consistent with H2, self-awareness had a negative influence on deception ($\beta = -.470$, $p < .01$). Taken together, the results support our prediction that avatar similarity (homophily) should lead to high self-awareness, which, in turn, reduces deception. Hence, H3 is supported.

With regards to appearance similarity, however, we found a significant direct path from appearance similarity to deception ($\beta = -.155$, $p < .05$). The association was negative, which suggests that individuals were less likely to deceive others when they perceived their avatars to be similar to themselves in terms of appearance.

Consistent with H4, homophily had positive associations with all three sub-dimensions of attraction, such as physical ($\beta = .244$, $p < .001$), social ($\beta = .568$, $p < .001$), and task ($\beta = .648$, $p < .001$). Hence, H4 is partially supported. As for the path between the three

dimensions of attraction and deception, only task attraction had a marginally significant association with deception ($\beta = -.754$, $p = .061$). Hence, H5 and H6 are not supported.

To summarize, we found (1) people who have greater homophily with their avatars are likely to have higher self-awareness, which, in turn, leads them to deceive less in their interactions; (2) these individuals are likely to have a strong attraction for their avatars on all three dimensions of physical, social and task attraction; (3) similarity in terms of appearance has a direct impact on deception, leading to less deception.

5. Discussion

The primary purpose of this paper was to study and understand deceptive communication in an avatar-mediated virtual environment by exploring the theoretical relationships between avatar use, self-awareness, similarity attraction, and deception. In short, the findings generally confirm our proposition that avatar similarity (homophily) affects the veracity of communication through self-focused attention.

First, we found that avatar similarity in attitudes and belief (i.e. homophily) reduces deception through its impact on self-awareness. Since homophily implies a close tie between avatar and user, the manner in which the user behaves and interacts through the avatar would be closely related to his or her beliefs, attitudes and emotions. Compared to an avatar that is not homophilous, the user would more likely turn his or her attention inwards when deciding on the behavior and response of the avatar since it is essentially mirroring his or her emotions and values. This inward focus of attention (i.e. self-awareness) makes salient internal standards (e.g., honesty), which, in turn, can lead people to behave in more normative ways, like tell the truth, so as to align their actions with standards. The findings show that avatars can affect behav-

ioral outcomes through their effect on self-awareness and have a role to play in fostering close relationships in virtual environments.

Second, we found that greater homophily leads to high perceptions of avatar attractiveness in physical, social and task aspects, bearing out our hypothesis on the similarity–attraction association. People who project their attitude, values and behavior fully on their avatars are expected to have greater attraction for it because of the shared attribute between the person and the avatar (Bailenson et al., 2008). However, none of the three dimensions of attraction displayed a significant association with deception, although the association between task attraction and deception approached marginal significance. It may be that attraction affects deception indirectly through other mediating variables (e.g., self-esteem, self-confidence) that were not included in this study. Future studies should examine these variables to determine if they have a mediating role in the relationship between attraction and deception.

Third, although it was not originally predicted, we found that physical similarity has a direct negative impact on deception. The hypothesized indirect effect of appearance similarity on deception via self-awareness was not supported as the link between appearance similarity and self-awareness was not significant. This is interesting, given that self-awareness is generally associated with physical similarity as with one's encounter with one's own image in the mirror. We think that although theoretically, self-awareness is heightened with the use of a mirror, the fact that the camera is behind and above the avatar means that exposure to the avatar's face and features is limited and this could have affected the degree of self-awareness. Furthermore, the creation of a physically similar avatar may inadvertently ground or reduce any fantastical imaginings or illusory projections one may be apt to conjure. Boellstorff (2008), in his ethnographic account, for instance, noted that “when residents referred to alts as ‘costumes’ or ‘masks,’ or emphasized that they created alts that did not sound or look ‘like themselves,’ the implication was that the avatar in question was ‘alternative’ to their primary avatar as much as their actual-world self” (p. 133). Appearance similarity between the user and avatar may reduce this feeling of being “masked”, and thereby negatively affect levels of deception.

Taken together, the findings suggest that self-awareness is a key factor that determines deception in virtual environments. When triggered by homophily, it has a significant impact on deception ($\beta = -.470, p < .01$). In contrast, the direct effect of physical similarity on deception is weaker ($\beta = -.155, p < .05$).

Overall, these findings highlight the value of this study: The strength of effect of avatar similarity on deception can be revealed only when (1) the two dimensions of similarity are separately examined, and (2) important mediating factors, such as self-awareness, are included in the research model, thereby revealing the strong impact of homophily on deception.

The study's findings have a useful practical application in using avatars to help enrich interpersonal communication through the creation of more truthful environments that discourage deception, and encourage and support interpersonal bonds. Health intervention programs and the military can also benefit from having participants in their game therapy or simulated environments be more frank about their problems or experience.

6. Limitations and directions for future studies

The current study has some limitations that also indicate directions for future research. The survey dealt with the sensitive topic of deception which could have made respondents less than truthful in their responses due to impression management reasons. Efforts were made to minimize any identifying information; few questions

asked for offline information and in the survey brief, participants were assured of the confidentiality of their input. Nonetheless, as avatar names were requested for payment purposes, some respondents may still have felt uneasy being truthful in their responses.

We collected the data for this study using convenience sampling. Although it was an easy and fast way to gather information, it may have threatened external validity. Given that it was impossible to obtain a comprehensive list of all Second Life users, random sampling was not a viable option for this type of research. Future studies, however, may try to draw a more generalizable sample to enhance external validity.

This study utilized self-report measures of perceived similarity. As mentioned earlier, we focused on perceived similarity rather than actual similarity since the former has been found to have a stronger effect than the latter in previous studies (Selfhout et al., 2009; Turban & Jones, 1988). However, as perceived similarity is based on perceptions, it is subjective. Future work could examine a more objective measure of similarity (i.e., actual similarity) by obtaining an independent assessment of similarity through the use of judges. This can help determine if the findings parallel that of perceived similarity, and ascertain outcomes arising from the differences.

Since appearance has a direct association with deception, future studies could examine the mechanisms that underlie this process and the presence of any mediating factors. In addition, the ways in which more malleable aspects of appearance (e.g., hair, eye color, clothes and skin color) and less often tweaked aspects (e.g., height, foot size and body shape) differ in influencing deceptive behavior can be examined. This would allow for a fuller picture of the impact of the various components of appearance on behavior.

Building on this study, future work could also include other variables, like sociability, to assess their effect on different communicative behaviors. Different types of avatars could also be examined to determine if certain behaviors are linked more strongly with a particular type of avatar. Experimental design can be used to have clearer evidence of causal relationships between avatar use and deception. For instance, less anthropomorphic avatars could be matched up against anthropomorphic ones to determine if they deceive differently.

7. Conclusion

Although numerous studies have been conducted to examine how avatars affect behavior, very little research has attempted to explore how avatars can affect the veracity of communicative behavior. This study has contributed to the existing research by showing the influence of avatars on deceptive behavior and the role of self-awareness in the process. Increasingly, as virtual environments employ avatars, there is a greater need to better understand the effect of avatars on communication. An important task is to specify underlying processes involved in avatar effect. We hope to see continued and more effective efforts aimed at addressing these crucial issues.

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